Rayen Chemlali

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SUMMARY

Currently a student at INP Toulouse – ENSEEIHT (N7) on an academic exchange, I am completing the final year of my engineering degree, majoring in HPC & Big Data, alongside a Master's degree in the PSMSC track (Scientific Computing). I come from a 4-year Software Engineering program at INSAT (National Institute of Applied Science and Technology) in Tunisia. Passionate about mathematics and computer science, I have taken part over the course of my studies in several programming contests and hackathons, alongside gaining professional experience through internships, freelance work and part-time roles.

KEY SKILLS

- **Generative & Agentic AI:** NLP, LLMs (prompt engineering, fine-tuning, embeddings), RAG, Agentic AI, multi-agent architecture, tool calling, LangChain, LangGraph, Qdrant
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, CNN, Transformer Architecture, time series, audio classification
- **Scientific Computing & HPC:** parallel and distributed computing, HPC algorithms, OpenMP, MPI, CUDA, OpenCL
- **Optimization:** Operations research (LP, MILP), heuristic search algorithms (A*), Gurobi
- **Big Data:** Hadoop, Spark, Kafka
- **Programming Languages:** Python, C++, Java, Go, JavaScript, PHP
- **Frameworks:** FastAPI, NestJS, ASP.NET, Symfony, React, Next.js, Angular
- **Software Development & Tools:** Software architecture, Design Patterns, Relational and non-relational databases, Git, Docker, Kubernetes, CI/CD, DevOps

PROFESSIONAL EXPERIENCE

Plug & Tel

ML Intern *Jun 2026 – Aug 2026*

Reduced the classification error rate of a production AI model for answering machine detection (AMD), while keeping latency compatible with the real-time constraints of VoIP streams. Explored several approaches: tabular models, CNNs on mel-spectrograms, and hybrid models combining both. Implemented model explainability: Grad-CAM for CNNs and tabular model explainability.

Part-Time Software Engineer *Nov 2025 – Apr 2026*

Worked as a part-time software engineer, contributing to software development projects and applying technical skills in a professional setting, notably developing VoIP microservices in Go and taking part in the production deployment process.

Elevvo Pathways

ML Intern *Mar 2026*

Completed 5 end-to-end ML projects covering clustering, classification, time series forecasting, computer vision and audio classification. Mastered the full ML pipeline, from data preprocessing to model evaluation.

Code

Talan Tunisie Consulting

Summer Intern *Jun 2025 – Aug 2025*

Designed and developed the first phase of the Smart Container Registry to optimize Docker storage costs, including a conversational chatbot with dynamic tool calling, a unified Kubernetes/S3 dashboard and custom deletion rules.

Frontend *Backend* *Demo*

Freelance

Software Developer *Various dates*

Worked on two freelance projects:

- **Strategy Copilot**: AI-powered business strategy copilot with a chatbot and graphical visualizations, generating diagnostics, financial projections and market analyses with PDF exports.
- **Shannon**: Co-developed a C++ code generator for electronic boards for the company Shannon, based on graph modeling, with a PyQt5 desktop application and an interactive Simulink-inspired canvas. *Demo*

PROJECTS

- **Multi-Agent Travel Planning System** Jan 2026 – Jun 2026
  End-of-year project: personalized travel planning system driven by multiple AI agents with conversational interaction, RAG and progressive knowledge base construction. Multi-level persistent memory and optimized response time through background pre-computation; evaluated with automated tests simulating users with different personas.
  LLMs, LangChain, LangGraph, RAG, Qdrant *Demo* *Report (full architecture details)*

- **Smart City Platform** Jan 2026 – Jun 2026
  Urban intelligence platform implementing a full Big Data pipeline combining weekly batch and real-time streaming on 6 real-world datasets (traffic, energy, crime). Computes via MLlib the weights of the City Pulse Score, a composite indicator per district for continuous urban monitoring.
  Hadoop, Spark, Kafka, MLlib

- **SoundGuard (Samsung Innovation Campus)** Oct 2025 – Nov 2025
  AI program capstone project: real-time detection of dangerous noise levels to protect workers' hearing; the system classifies audio signals and triggers alerts when exposure exceeds safety thresholds.
  Demo *Code*

- **MediSys** Mar 2025 – Jun 2025
  Medical management platform covering consultations, appointments, prescriptions and lab tests. Automatic digitization of medical documents via OCR. Includes a conversational chatbot that recommends nearby doctors and medications tailored to the patient's case.
  NestJS, Next.js, GraphQL, Python, LLMs, OCR *Frontend* *Backend* *Demo*

- **Heterogeneous GPU Computing (CUDA/OpenCL)** Apr – May 2025
  Two projects completed within the same course: dynamic load balancing across CPU, iGPU and dGPU in OpenCL, achieving a 1.23× speedup over the GPU alone; and CUDA kernel optimization (tiling, float4 vectorization, Tensor-Cores), reaching 1235 GFLOPS, within 6% of cuBLAS performance.
  CUDA, OpenCL, cuBLAS, C++ *Reports*

- **Operations Research Optimization** Apr – May 2025
  Modeling and solving three NP-hard problems with Gurobi (transportation, VRP with MTZ subtour elimination, facility location), including dual and sensitivity analysis. Delivered two complete applications: a Next.js/FastAPI web platform with interactive route visualization, and a PyQt5 desktop application.
  Python, Gurobi, Next.js, FastAPI *Code* *Report*

EDUCATION

INP Toulouse – ENSEEIHT *2026 – 2027*
– 5th year, HPC Big Data track (academic exchange)
– Master's degree, PSMSC track (Scientific Computing)

National Institute of Applied Science and Technology (INSAT) *2022 – 2026*
1st to 4th year in Software Engineering

Baccalaureate in Mathematics, Tunisia *2022*
Highest Honors

LANGUAGES

Arabic – Native **French** – Fluent **English** – Fluent **Spanish** – Beginner

CERTIFICATIONS

- Samsung Innovation Campus AI Program
- Efficient LLM Customization (NVIDIA)
- Building RAG Agents with LLMs (NVIDIA)

ACHIEVEMENTS

- Thematic winner (Enedis track) – AI For Impact Hackathon by La Mêlée, Oct 2026
- Top 6 – Data Quest 2026, Feb 2026 *Code* *Demo*
- 3rd place – AI 4 New Generation Hackathon, EY & Univ. Ibn Khaldoun, 2024
- Bronze medal (11th) – TCPC 2024
- 2nd place – Code Quest 2023, INSAT